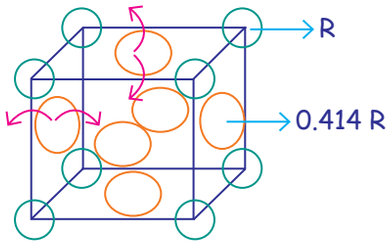
11 Solid State

SOLUTION (QR CODE)

10. $(\frac{\pi}{6}[1+3(0.414)^3])$



Effective no. of atoms of radius R is equal to 1.

Effective no. of atoms for radius (r) = 0.414 R = 3

$$PF = \frac{1\times\frac{4}{3}\pi R^3 + 3\times\frac{4}{3}\pi(0.414R)^3}{a^3 = (2R)^3}$$

$$= \frac{\frac{4}{3}\pi R^3\left[1+3(0.414)^3\right]}{8R^3}$$

$$= \frac{\pi}{6}[1+3(0.414)^3]$$

12. (D) $4R = a\sqrt{2}$ so, $a = 2\sqrt{2}R$

Area of square unit cell = $(2\sqrt{2}R)^2 = 8R^2$

Area of atoms present in one unit cell

$\pi R^2 + r\left(\frac{\pi R^2}{4}\right) = 2\pi R^2 = 2\pi R^2$ so, packing efficiency

$$= \frac{2\pi R^2}{8R^2}\times 100 = (\pi)(4)\times 100 = 78.54\%$$

14. 0.732 R = 75, $R = \frac{75}{0.732}$ pm

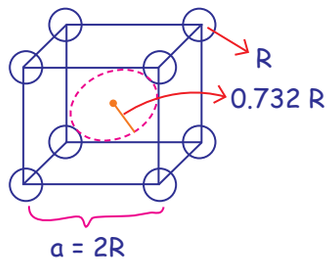
$$a = 2R = 2\left(\frac{75}{0.732}\right)\text{pm}$$

Next to next to next nearest atom distance is 2a.

$$2a = 4\left(\frac{75}{0.732}\right)\text{pm} = \frac{300}{0.732}\text{pm} = 409.8 \text{ pm}$$

Are shabdo ko samjho pehle, sawaal to tabhi hoga

16. (C)



$$P.F = \frac{\text{volume of effective atoms}}{\text{volume of unit cell}}$$

$$= \frac{1\times\frac{4}{3}\pi R^3 + 1\times\frac{4}{3}\pi(0.732R)^3}{a^3 = (2R)^3} = 0.73$$

Muskil nhi hai bachcho

21. Calculate unit cells of radius 0.414 R

$$= 12\left(\frac{1}{4}\right) + 1 = 3 + 1 = 4$$

$$\frac{P.F = 4\times\frac{4}{3}\pi R^3 + 4\times\frac{4}{3}\pi(0.414R)^3}{a^3 = \left(\frac{4R}{\sqrt{2}}\right)^3}$$

$$= \frac{\frac{16\pi}{3}\cancel{R^3}\left[1+(0.414)^3\right]}{\frac{32}{\sqrt{2}}\cancel{R^3}} = \frac{\pi\sqrt{2}}{6}\left[1+(0.414)^3\right]$$

22. Next^2 $\Rightarrow a\sqrt{\frac{3}{2}} = 150$, $a_{FCC} = 150\sqrt{\frac{2}{3}}$ pm

$a_{FCC}\sqrt{2} = 4R$

$$R = \frac{a_{FCC}\sqrt{2}}{4} = \frac{150\sqrt{2}\sqrt{2}}{\sqrt{3}\times 4} = \frac{75\sqrt{3}}{3} = 25\sqrt{3}\,\text{pm}$$

size of the atom which can be fit = 2(0.414R)

$$= 2\left(0.414 \times 25\sqrt{3}\right) = 38.85\,\text{pm}$$

Yeh dimag ko pura ghumaega

24. For the maximum diameter of the atom in the interstitial site, octahedral void must be taken because they can accommodate a larger atom.

$$\frac{r}{R} = 0.414$$

For FCC, $R = \frac{a}{2\sqrt{2}} = \frac{400}{2\sqrt{2}}$, $\frac{r}{R} = 0.414$

$$\Rightarrow r = 0.414 \times \frac{400}{2\sqrt{2}} = 58.55 \text{ pm}$$

Diameter is 2r = 2 × 58.55 = 117.1 pm

28. (B) Packing fraction $= \dfrac{3 \times \frac{4}{3}\pi r^3 + 1 \times \frac{4}{3}\pi r_-^3}{a^3}$

$$= \frac{\frac{4}{3}\pi\left[3\left(\frac{r_+}{r_-}\right)^3 + 1\right]}{\left(\frac{a}{r_-}\right)^3}$$

$2r_- = a$ & $\frac{r_+}{r_-} = 0.414$

∴ Packing fraction = 0.634

30. Density = $\frac{\text{Mass}}{\text{Volume}} = \frac{\text{Effective atoms} \times M}{a^3 \times N_A}$

$$2 = \frac{\text{Effective atoms} \times 75}{(5\times10^{-8})^3 \times 6 \times 10^{23}}$$

Effective atom in one unit = 2 (BCC)

Sedha formula lagega bus!

For bcc lattice diagonal of the unit cell

$$d = 2r$$

$$= \frac{\sqrt{3}a}{2} = \frac{4 \times \text{Atomic radius}}{\sqrt{3} \times \text{edge length}}$$

$$\Rightarrow 4r = \sqrt{3}a$$

$$\Rightarrow 4r = \sqrt{3} \times 5.0 \times 10^{-10}\text{m}$$

$$r = \frac{\sqrt{3}}{4} \times 5.0 \times 10^{-10}\text{m}, \; r = 216.5 \text{ pm}$$

32. Wt. of 1 cm^3 (1 ml) is $\frac{4}{\pi}$ gm

Moles of atoms in 1 cm^3 volume = $\frac{4}{\pi} \times \frac{1}{40}$

$= \frac{1}{10\pi}$ mol

No. of atoms of Ar = $\frac{1}{10\pi} \times 6 \times 10^{23}$

Volume of Ar atoms = $\frac{1}{10\pi} \times 6 \times 10^{23} \times \frac{4}{3}\pi$ $(1.5 \times 10^{-8})^3$

$$\text{P.F} = \frac{\text{Volume of atoms}}{\text{Volume of unit}}$$

$$= \frac{\frac{1}{10\pi} \times 6 \times 10^{23} \times \frac{4}{3}\pi(1.5\times10^{-8})^3}{1} = 0.27$$

$$\boxed{\text{V.F} = 0.73 = 73\%}$$

33. (A) $\frac{\text{P.F}}{\text{density}} = \dfrac{\frac{\text{Volume of atoms}}{\text{Volume of unit}}}{\frac{\text{Mass of unit}}{\text{Volume of unit}}}$

$= \frac{\text{Volume of atoms}}{\text{Mass of unit}}$ (1 Å = 10^{-8} cm)

$$0.1 = \frac{Z \times \frac{4}{3}\pi(10^{-8})^3}{Z \times \frac{M}{6\times10^{23}}}, \; M = 8\pi$$

36. $h = 4r\sqrt{\frac{2}{3}}$

$\frac{h}{2} = 2r\sqrt{\frac{2}{3}}$ → distance between two consecutive closed packed layer.

$8 \times 2R\sqrt{\frac{2}{3}} = 800$ $\quad a_{BCC} = \frac{4R}{\sqrt{3}} = \frac{4}{\sqrt{3}}50\sqrt{\frac{3}{2}}$

BCC बनाया

$R = 50\sqrt{\frac{3}{2}}$ pm $\quad = \frac{200}{\sqrt{2}} = 100\sqrt{2}$

next to next = $a_{BCC}\sqrt{2}$

$= 100\sqrt{2}\,\sqrt{2}$ = (200 pm)

38. To find empty space, first find filled space and then subtract filled space from total space.

The following is the set-up: ABABAB..... stands for hexagonal close-packed unit cell with six atoms per unit cell. Additionally, the volume of the unit cell $= 24\sqrt{2}r^3$

$\Rightarrow$ Packing fraction

$$= \frac{\text{Volume occupied by atoms}}{\text{Volume of unit cell}}$$

$$= 6 \times \frac{4}{3}(3.14)r^3 \times \frac{1}{24\sqrt{2}r^3} = 0.74$$

$\Rightarrow$ Percent empty space = 100(1 – 0.74) = 26%.

46. (A) $\therefore$ No. of octahedral voids = 4 and No. of tetrahedral voids = 8

As m fraction of octahedral voids is occupied by Al^{3+} ions, therefore, Al^{3+} ions present = 4m. Similarly, Mg^{2+} ions = 8 n. Hence, formula of the mineral is $Al_{4m}Mg_{8n}O_4$. As total charge on the compound is zero, hence

4m(+3) + 8 n(+2) + 4(–2) = 0

or 12 m + 16 n – 8 = 0

Substituting the given values of m and n, equation is satisfied only when m = $\frac{1}{2}$ and n = $\frac{1}{8}$

52. $4r_{Cl^-} = \sqrt{2}a$

$$r_{Cl^-} = \frac{\sqrt{2}a}{4}$$

$$r_{Cl^-} = \frac{\sqrt{2} \times 5.14Å}{4}$$

$$r_{Cl^-} = 1.82\ Å$$

Now, the radius ratio for NaCl structure is 0.414 – 0.732 hence, LiCl will have the same radius ratio. So, radius of Li^+ can be calculated as

$r_{Li^+} = 0.414 \times r_{Cl^-}$	$r_{Li^+} = 0.732 \times r_{Cl^-}$
$r_{Li^+} = 0.414 \times 1.82\ Å$	$r_{Li^+} = 0.732 \times 1.82\ Å$
$r_{Li^+} = 0.754\ Å$	$r_{Li^+} = 1.332\ Å$

Hence, the radius of Cl^- ion = 1.82 Å and the radius if Li^+ ion will be in the range of 0.754 Å – 1.332 Å.

58. (i) In rock-salt like crystal AB, there are four AB units per unit cell. Therefore, density (d) is

$$d = \frac{4 \times 6.023\ y}{6.023 \times 10^{23} \times 10^{-27} \times 8y}$$

$[\because a = 2y^{1/3}\ mm = 2y^{1/3} \times 10^{-9}\ m]$

$= 5 \times 10^3\ g/m^3 = 5\ kg/m^3$

(ii) Because the observed density exceeds the calculated density, there must be some extra metal in the interstitial gaps. Metal excess defect is the term for this type of flaw.

59. (B,C)

(B) The smaller atom or ion is dislodged from its regular lattice locations and encompasses the interstitial space in a Frenkel defect.

(C) Correct Proposition. Some anions depart the lattice in the F-centre defect, while the unoccupied sites trapped the electrons, maintaining the solid's overall electroneutrality.

62. Frenkel defects involve the movement of an ion from its lattice site to an interstitial site, creating a vacancy at the original site. This can lead to the formation of impurity energy levels within the bandgap. As a result, the bandgap energy of the crystal may decrease due to the introduction of these impurity energy levels.